Date: 12th- June- 26

AI INTERNSHIP TASK

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1.2 Math-for-ML Primer

Learning objective: Build just enough math intuition to *read* and *reason about* ML — not to pass an exam. You need to recognize the notation and understand what these operations represent, because they're the vocabulary of every algorithm you'll meet.

T1.2a — Math concepts explainer (in your own words)

Cover each of these with a 2–4 sentence plain-language explanation + a tiny worked example:

14. **Vectors and matrices** — what they are; do one 2×2 matrix multiplication by hand, showing the steps.

    1. vectors:
       list of numbers, or we can also refer to them as one dimensional array.
    2. matrices:
       rows and columns of a list of numbers, they are also called 2D arrays.

Similarities:

- they're the building blocks of data
- they both rely on numpy for data manipulation, analyzing and more
- they are used since computers can't read texts, it needs numbers and a collection of numbers are used to represent data like images

2x2 matrix multiplication method:

For 2x2 dot multiplication between matrices we use the dot multiplication (@) which results in the multiplication of arr1(rows) x arr2(columns). If we the following matrices for example

```
arr1 = np.array([
                [ 2 , 3 ],
                [ 3 , 5 ]
               ])
```

```
arr2 = np.array([
                [ 1 , 0 ],
                [ 4 , 5 ]
               ])
```

Steps	Result
It'll be arr1 @ arr2= • (1,1) = (2*1) + (3*4) = 2+12 = 14 • (1,2) = (2*0) + (3*5) = 0+15 = 15 • (2,1) = (3*1) + (5*4) = 3+20 = 23 • (2,2) = (3*0) + (5*5) = 0=25 = 25	output: [14 , 15] [23 , 25]

15. **Dot product** — compute the dot product of `[1, 2, 3]` and `[4, 5, 6]` by hand (answer: `1·4 + 2·5 + 3·6 = 32`). Explain what a dot product *tells* you (a measure of how aligned two vectors are).

- Vector dot multiplication is easier but they should be equal in length. For example
- vector1 = np.array([1, 2, 3])
- vector2 = np.array([4, 5, 6])
- vector1 @ vector2 =
- (1*4) + (2*5) + (3*6) = 32
- This helps in finding how much vectors/arrays align or agree with each other, which helps the neural networks (core of Deep learning) to make decisions, for example answering questions like "would this be a good ad for this person?"

16. **Derivative / gradient** — explain what a derivative represents (rate of change / slope) and what a gradient is (the direction of steepest increase). This is the heart of how models learn.
- <u>Derivative:</u>
  Represents the rate of change in the output with respect to some variable ("How does it affect the result if i change the value of a certain input?, does the error decrease or increase?"). It is represented as a slope in the graph where we aim for the lowest point/valley (least error)
- <u>Gradient:</u>
  Multiple derivatives are called gradient, where instead of comparing a single variable we have many. It acts like the path we should be going to reach the valley (least error), where it shows how the model is performing or how bad it is currently, it points to what is bad, but not how to fix it

17. **Mean, variance, standard deviation** — compute all three for `[2, 4, 4, 4, 5, 5, 7, 9]` by hand.

<table>
<tr><th>mean</th><th>(n - mean)</th><th>distance**2</th><th>variance</th><th>STD</th></tr>
<tr><td rowspan="3"><u>Mean:</u> the average value, add all values and divide them with their count</td><td>2 - 5 = -3</td><td>-3**2 = 9</td><td rowspan="3">Variance: squared STD, represents an outlier or extreme value</td><td rowspan="3">Standard deviation: measures the spread of data from the average (mean)</td></tr>
<tr><td>4 - 5 = -1</td><td>-1**2 = 1</td></tr>
<tr><td>4 - 5 = -1</td><td>-1**2 = 1</td></tr>
<tr><td rowspan="5">2+4+4+4+5+5+7 +9) / 8 =

40/8 =

5</td><td>4 - 5 = -1</td><td>-1**2 = 1</td><td rowspan="5">9+1+1+1+0+0+4 +16 = 32

32/8 = 4</td><td rowspan="5">sqrt(4) =
2</td></tr>
<tr><td>5 - 5 = 0</td><td>0**2 = 0</td></tr>
<tr><td>5 - 5 = 0</td><td>0**2 = 0</td></tr>
<tr><td>7 - 5 = 2</td><td>2**2 = 4</td></tr>
<tr><td>9 - 5 = 4</td><td>4**2 = 16</td></tr>
</table>

18. **Probability basics** — explain conditional probability, then solve this classic problem with arithmetic shown:

> A disease affects 1% of people. A test is 99% accurate (99% true positive, 99% true negative). You test positive. What's the actual probability you have the disease? *(The surprising answer is ~50% — this is Bayes' theorem, and understanding why is the point.)*

Conditional Probability: counts the percentage/likelihood of an event occurring given that a condition is valid too, for example ("what's the probability of a biology exam tomorrow IF you have a physics exam too?")

solution:

let's take a certain **population** in mind, for example **100,000**:

the disease infects 1% = **0.01**

- **diseased(D):** 100,000 * 0.01 = **1000**
- **healthy(H):** 100,000 - 1000 = **99,000**

test:

- **TP: D * TP =** 1000 * 0.99 = **990**
- **TN: D * TN =** 1000 * 0.99 = **990**
- **FP: D - TP =** 1000 - 990 = **10**
- **FN: D - TN =** 1000 - 990 = **10**
- **total T =** 990 + 990 = **1980**
- **PROBABILITY OF TP WHEN T:** TP/T = 990/1980 = **0.5**

Constraint: Plain language a non-mathematician could follow. No copied definitions — your own words.

Target: Mentor reads it and agrees you understand each concept (not just restated a textbook).

Deliverable: A 2–3 page document (Markdown or PDF) with explanations and worked examples.